

ASHOK GURUNG, Ph.D.

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Professional Experience

Postdoctoral Research Associate Department of Mechanical Engineering Rochester Institute of Technology, NY, USA	Sept 2024 - Present
Quantum Field Expert and Outreach Specialist Department of Mechanical Engineering Rochester Institute of Technology, NY, USA	Sept 2024 - Nov 2024
Research Assistant Department of Physics University of Connecticut, CT, USA	Dec 2020 - Aug 2024
Teaching Assistant Department of Physics University of Connecticut, CT, USA	Aug 2018 - Dec 2020
Teaching Assistant Department of Physics Wright State University, OH, USA	Aug 2016 - July 2018

Education

University of Connecticut (UConn), CT, US	Ph.D. in Physics, December 2024
Wright State University (WSU), OH, US	M.S. in Physics, July 2018

Teaching Experience

NSF-Engine Workforce Development Summer Camp <i>Postdoctoral Research Associate, Department of Mechanical Engineering, RIT</i>	Apr 2025 – May 2025 <i>Rochester, NY</i>
• Participated in curriculum development for the 2025 NSF-Engine Workforce Development Summer Camp for community college students. • Will deliver a lecture on galvanic cells and their components, adapting content for accessibility to students without prior technical background. • Developed instructional materials explaining battery classification (based on life cycle, format, electrode architecture, etc.) and operational principles. • Collaborated with graduate students Radwa Elawadly and Imtiaz Shozib to integrate lecture content with hands-on experiments, ensuring theoretical concepts were reinforced through experiential learning.	
Quantum Field Expert and Outreach Specialist <i>Postdoctoral Research Associate, Department of Mechanical Engineering, RIT</i>	Sep 2024 – Nov 2024 <i>Rochester, NY</i>
• Contributed as a quantum field expert at the event titled "Lecture: What Are Quantum Technologies?" , organized by Quantum CT, a collaborative initiative between the University of Connecticut and Yale University. • Played a key role in promoting quantum technologies and their impact on economic growth and societal advancement. • Prepared and tailored content to educate diverse audiences, including educators, businesses, policymakers, and engineers, on the importance of quantum technologies and their future impact. • For more details, visit: https://quantumct.org/events/quantum-up-challenge/panel-sessions/	

Studio Lab Teaching Assistant

Jan 2019 – Dec 2020

Department of Physics, UConn

Storrs, CT

- Taught introductory physics lab courses: PHYS 1501, and 1502 for both physical and non-major (social sciences and arts, Computer Science, Business, Nursing) students.
- Delivered live demonstrations to illustrate key physics principles, enhancing student understanding through hands-on learning.
- Prepared and conducted numerical problem solving sessions at the end of lectures to reinforce conceptual understanding.
- Facilitated group discussions, guiding students through interactive problem-solving with their peers.
- Committed to teaching excellence at the undergraduate level by offering flexible office hours for personalized discussions, ensuring clarity in complex concepts.
- Assisted instructors during class sessions, gathering direct student feedback to help refine lecture delivery and content.
- Collaborated with the lab manager and course instructor to design experiments, structure course content, and develop numerical examples.
- Assisted in standardizing grading rubrics for exams grades to ensure fair and consistent evaluation.

Traditional Lab Teaching Assistant

Aug 2016 – Dec 2018

Department of Physics, UConn and Wright State University

- Taught both calculus- and algebra-based physics labs, including PHYS 1201, 1202, 1401, 1402, 1501, and 1502, engaging students with hands-on experiments to reinforce theoretical concepts.
- Designed and structured lab sessions to closely align with lecture content, ensuring a seamless learning experience for students.
- Actively sought anonymous student feedback to continuously refine teaching methods and enhance student engagement.
- Maintained regular communication with instructors to synchronize lab activities with lecture topics, creating a cohesive learning environment.

Student Mentorship

- Alhamdu Nuhu Bage – Ph. D. student (RIT)
- Dan Brunner – Master student (RIT)
- Charles Schwarz – Bachelor Student (UConn)
- Islam Bilalovski – Bachelor Student (UConn)

Publications

- **A. Gurung**, M. Fatin Ishtiyag, T. Sahoo, S. M. Nakhmanson, S. K. Nayak, S. P. Alpay, and P. E. Janolin, “*Bioelectrostriction: An Invaluable Principle for Bioimaging and Diagnostics*,” (in preparation 2024).
- **A. Gurung**, J. M. Mangeri, M. Fatin Ishtiyag, S. P. Alpay, and S. Nakhmanson, “*Extrinsic Dielectric Response due to Domain Wall Motion in Ferroelectric BaTiO₃*,” Computational Materials Today. 5, 100016 (2025).
- **A. Gurung**, J. M. Mangeri, A. M. Hagerstrom, N. D. Orloff, S. P. Alpay, and S. Nakhmanson, “*Modeling Structure-Properties Relations in Compositionally Disordered Relaxor Dielectrics at the Nanoscale*,” J. Appl. Phys. 134, 104102 (2023).
- **A. Gurung***, D. Trujillo* (equal contribution), J. Yu, S. K. Nayak, S. P. Alpay, and P-E. Janolin, “*Data-driven Methods for Discovery of Next-generation Electrostrictive Materials*,” npj Comp. Mater. 8, 251 (2022).
- **A. Gurung**, “*Molecular Dynamics Study of Hydrogen Trapping and Helium Clustering in Tungsten*,” master’s thesis, Wright State University, 2018.

Research Experience

Multiphysics Modeling

Postdoctoral Research Associate, Department of Mechanical Engineer, RIT

Aug 2024 – Present

Rochester, NY

Mesoscopic and Continuum Scale Modeling

Research Assistant–Funded by Air Force Research Lab (AFRL) and Industry

Dec 2020 – Aug 2024

Storrs, CT

Atomistic modeling–(Density Functional Theory)

Research Assistant, Physics Department, UConn

Aug 2018 – Dec 2020

Storrs, CT

Atomistic Modeling–(Classical Molecular Dynamics)

Research Assistant, Physics Department, WSU

Aug 2016 – July 2018

Dayton, OH

Presentations (Oral) and Conferences

- “*Modeling the dielectric dispersion of BaTiO₃ with a local phase field method*”, Materials Science and Technology Conference, October 2023, Columbus OH, USA
- “*Modeling the relaxor dielectric dispersion of Ba_{1-x}Sr_xTiO₃ with a local phase field method*”, Materials Science and Technology Conference, October 2022, Pittsburgh PA, USA
- “*Modeling the relaxor dielectric dispersion of with a local phase field method*”, Electronic Materials and Applications Conference, January 2022, Virtual participation, USA
- “*Predicting dielectric response in 5G materials with FEM simulations*”, Industrial affiliates program (IAP) annual meeting, July 2019, UConn, Storrs CT, USA.

Professional Memberships and Services

- Served as a **peer reviewer** for **Physical Review B**, **Physical Review Research** and **Physical Review Applied**
- American Physical Society
- American Ceramic Society (ACerS)
- Material Advantage
- Association for Iron and Steel Technology (AIST)

Workshops

- “*Moving ions with VASP*” Organized by VASP, Virtual, Nov 2023.
- “*MOOSE Workshop*” Organized by Idaho National Lab, Virtual, June 2023.
- “*Quantum Matter: Dynamics and Sensor*” jointly organized by the University of Connecticut, University of North Florida, and NORDITA, December 2022.
- “*CHiMaD Phase Field Workshop XIV*” organized by Center for Hierarchical Material Design, Evanston, IL, Hybrid, October 2022.
- “*CHiMaD Phase Field Workshop XII*” organized by Center for Hierarchical Material Design, Virtual, November 2021.
- “*IMS Quantum Matter Group 2019*” held at Los Alamos National Lab (LANL) as a visiting graduate student for the summer program 2019.

Honors Scholarships and Fellowships

- “*Modeling structure–properties relations in compositionally disordered relaxor dielectrics at the nanoscale*” paper **featured on the cover of Journal of Applied Physics** (2023).
- Conference Participation Award, UConn (2023).
- Doctoral Dissertation Fellowship, UConn (2023).
- Pre-Doctoral Dissertation Fellowship, UConn (2019).
- Guest Travel Fund, Los Alamos National Lab (LANL) (2019).

Project and Relevant Coursework

Graduate Level Coursework: Solid State Physics, Condensed Matter Physics, Statistical Mechanics, Mathematical Physics, Electrodynamics I and II, Quantum Mechanics I, II, III.

Graduate Level Computational Physics [Coursework]: Various **numerical solver**(Euler, Runge-Kutta, finite differences); **Krylov subspace** method(conjugate gradient, lanczos etc); Error analysis techniques (Von Neumann Stability, CFL condition, Round off error etc).

Phase Transformation in Solids [Project]: Developed an **open source app** based on FEM to predict the **thermal behavior** of various alloys under **phase transition**.

Programming [Udemy-Coursework]: Beginner to Advance—Deep Dive in C++, Machine Learning.

Technical Skills

Computational techniques: Finite Element Modeling (FEM); Classical Molecular Dynamics; Density Functional Theory (DFT) via VASP; Data Mining; Machine Learning

Programming and Software: Python; MATLAB; C/C++; shell script; MS Office; LaTeX; GitHub

Operating System: Ubuntu; high-performance computing cluster; and Windows OS

Other Skills: Interpersonal & Leadership Skills; Scientific Writing & Presentation Skills; Project Management & Organization Skills; Time Management Skill; Teamwork Skill; Independent Work Skill

Leadership

- Senate, Graduate Student Senate, (2021-2023)
- Secretary, Nepalese Graduate Student Association, (2021-2022)
- Member, Student Welfare Committee, (2022-2023)
- Member, Graduate Employee and Postdoc Union (2018-2023)

References

- Dr. Serge Nakhmanson, serge.nakhmanson@uconn.edu, Associate Professor, Department of Materials Science and Engineering, University of Connecticut, CT, USA
- Dr. Xian Wu, xian.wu@uconn.edu, Associate Professor In Residence, University of Connecticut, CT, USA
- Dr. Belter Ordaz Mendoza, belter.ordaz@uconn.edu, Associate Professor In Residence, University of Connecticut, CT, USA